FieldLENS → a Publishable XR Research Contribution

Deep-research report · prepared for Avilash Angirekula · June 2026

Bottom line up front

A defensible, publishable contribution exists for you — but it is **not** a systems/tech contribution (a new tracking or overlay-generation system). The literature shows that lane is already occupied by funded, multi-author teams. Your realistic and genuinely novel lane is a **rigorous controlled human-subjects evaluation in a safety-critical domain**: measuring how novices behave when AI-driven AR guidance is *imperfect* on lethal-voltage equipment. That is the one place where your HVAC / novice / real-stakes framing is a research *strength* instead of a generic demo, and the one place the current literature has a real hole.

Verification note. The research workflow ran 107 agents, extracted 115 claims, and adversarially verified 25 (3 independent skeptics each; kill on 2/3). 21 confirmed, 4 killed. Confidence is flagged inline; arXiv preprints are marked as such.

0. Deadlines (verified live)

Venue	Paper deadline	Conference	Status for you
IEEE VR 2027	Abstract Aug 24, 2026; full Aug 31, 2026	Feb 27 - Mar 3, 2027, Melbourne	Realistic full-paper / poster target
ISMAR 2026	Mar 16, 2026 (passed)	Oct 2026	Passed; posters may still be open
ISMAR 2027	Not posted (expect ~Mar 2027)	Oct 2027	Plausible later target
CHI 2027	Sep 10, 2026	2027	Strong fit; hardest reviewing
UIST 2026	Mar 31, 2026 (passed)	Nov 2026	Passed; UIST 2027 ~Mar 2027

The two live, actionable targets for a 6-month effort starting now are **IEEE VR 2027 (Aug 31, 2026)** and **CHI 2027 (Sep 10, 2026)**. Both have poster / Late-Breaking-Work tracks with later deadlines and a much lower bar — likely your realistic tier. Verify poster deadlines on each site closer to the date.

Sources: ieeevr.org/2027; ieeeismar.net/2026/call-for-papers; chi2027.acm.org/authors/papers; uist.acm.org/2026/cfp.

1. What IEEE VR / ISMAR actually accept

Across the five adjacent areas, accepted work clusters into two recognizable shapes: (i) a new sensing / tracking capability, or (ii) a rigorous study on real experts doing real tasks on real hardware. Knowing which you are writing is half the battle.

A. AR for industrial / maintenance / field-service

- **Henderson & Feiner, ISMAR 2009 (Best Paper)** — the canonical template. Within-subjects, professional military mechanics, 18 real tasks inside an armored-vehicle turret; head-worn AR localized tasks ~56% faster. *(confirmed)* This is why reviewers expect real experts + real equipment.

- **Mahmood et al., Sensors 2023, N=28 between-subjects** — HoloLens vs paper manual; AR cut high-demand task time 14.94% ($p<0.05$), measured with EEG + NASA-TLX. *(confirmed)*

Typical eval: N~20-48, within/between subjects, concrete baseline (paper/monitor), metrics = completion time + error count + NASA-TLX + sometimes physiological. Accepted vs rejected turns on ecological validity and a real baseline, not toy block-assembly.

B. AR procedural guidance & task assistance (crowded)

"AR overlay beats paper manual for assembly" is essentially a solved finding. **Belga et al., HFES 2025, N=48** (2x3x3 within-subjects, Black Hawk cockpit; time, mental effort, discomfort) is a strong recent example — and note their counter-intuitive cue result below. *(confirmed)*

C. AI / vision-driven AR (the frontier, and your trap)

- **GBOT, IEEE VR 2024** (Li, Schieber, et al.) — real-time markerless 6D pose for AR assembly guidance. *(confirmed)*
- **Guided Reality, UIST 2025** (Zhao, Gunturu, Do, Suzuki) — automatically generates 3D world-locked AR guidance from VLMs (GPT-4o + Gemini 2.5 + SAM 2.1), 5 overlay types incl. component highlight; N=16 within-subjects vs static-arrow baseline + 4 expert interviews. *(confirmed)*
- **JARVIS** (arXiv 2604.10108, preprint) — VLM just-in-time AR instructions; formative 4x3 within-subjects N=10; image guidance cut error 3% vs 13% text ($p=0.001$). *(confirmed)*

Hard truth (high confidence): FieldLENS's pipeline — VLM identifies a component, generates a highlight box + steps — is a *strict subset* of Guided Reality and JARVIS, and yours is not even world-locked. As a *systems* contribution, the current product is below the bar and would be desk-rejected as "already done, less capable."

D. AR safety & attention guidance (your opening)

- **Camacho-Fidalgo et al., 2025 (arXiv 2503.04075), N=60** — wearing an AR-HMD did *not* improve hazard recognition and significantly *increased* accident-proxy events vs a smartphone; authors conclude current AR-HMD tech is "not ready" for hazardous industrial settings. *(confirmed; form-factor-specific — see caveat)*

E. AR interaction / cueing techniques

- **Attention Funnel, Biocca et al., CHI 2006** — 3D cue for occluded targets: +65% search consistency, +22% speed, -18% workload. *(confirmed)*
- **Lu et al., TVCG 2013/14** — "subtle cueing"; explicit cues can degrade visual search by distorting the scene. *(confirmed, conditional)*
- **Seeliger et al., ISMAR-Adjunct 2021** and **Renner & Pfeiffer, IEEE 3DUI 2017** — eye-tracking as a rigorous DV: gaze distribution, duration, path. *(confirmed)*

Two killed claims (refuted by the verifiers, so not relied upon): that head-worn AR beat LCD instructions in one specific maintenance study (0-3), and that JARVIS's *main* study was N=14 (1-2). Don't cite those.

2. Where the real gaps are (honest)

Crowded / avoid:

- "AR procedural overlay vs paper manual improves time/errors" — done to death.
- Novel VLM-drives-AR-overlay systems — done (Guided Reality, JARVIS), and out of solo scope anyway.

- Multimodal AR sensing stacks (e.g. ARGUS, TVCG/IEEE VIS 2023, ~18 authors, multi-year). Unrealistic solo.

Underexplored / where your domain is a strength:

- **Safety-critical evaluation of AI-driven guidance.** The N=60 result is a citable open problem: in hazardous settings AR can fail to help or actively hurt. Your lethal-voltage / true-novice framing is exactly that setting.
- **Over-reliance on *imperfect* AI in AR.** Every AI-AR guidance paper assumes the AI is right. Yours is provably ~70-80% accurate and “too lax on intact panels.” Nobody has safety-grounded what happens when a novice trusts a *wrong* AR safety verdict. This is the most novel angle in the report.
- **Cue design for cluttered/occluded panels.** Belga found simple cues beat complex ones in dense environments, contradicting older results. HVAC panels are dense and occluded — a cue comparison with hazard-fixation as the DV is a clean, novel question.

3. Project ideas (my analysis of FieldLENS)

★ Idea 1 - “Misplaced Trust”: novice over-reliance on imperfect AI safety verdicts in AR

- **(a) Claim)** When AI-driven AR guidance gives a *wrong* safety verdict on electrical equipment, novices over-rely on it and miss the error — and a design intervention (confidence display / verify-this cue / forced manual-check step) measurably reduces that over-reliance.
- **(b) Novelty)** Guided Reality / JARVIS assume correct AI. Automation-bias work exists in dashboards/decision-aids but not safety-grounded in AR field guidance with a real imperfect VLM. You'd be first where the AI's error rate is real and the stakes are lethal.
- **(c) Eval)** Controlled within-subjects, novices, FieldLENS-style overlays on photos/mockups of de-energized HVAC. Inject known wrong verdicts on a subset. DVs: hazard-detection rate, false-“safe” acceptance rate, trust, NASA-TLX, time. Compare verdict-only vs verdict+confidence vs verdict+forced-check.
- **(d) Scope)** Realistic. No new tracking. Hardest part is recruiting + IRB, not engineering.
- **(e) Venue)** CHI (best fit) or IEEE VR/ISMAR poster; full paper possible at a second-tier HCI venue.
- **(f) Hardware)** None required beyond phone/laptop. Eye-tracking optional.
- **(g) Startup)** Maximal — your “model is too lax / ~70-80%” problem becomes the research question, and the intervention is a feature you should ship anyway.

Idea 2 - Cue-design comparison for occluded HVAC panels

- **(a))** An attention-funnel / subtle cue redirects novice gaze to correct and hazardous regions better than an explicit box, without raising workload.
- **(b))** Closest: Biocca (attention funnel), Lu (subtle cueing), Belga (dense-env simple cues). Differentiator: hazard fixation as DV + true novices + electrical domain.
- **(c))** Within-subjects, 3 cue types x cluttered panel images; eye-tracking recommended (gaze-to-hazard latency, fixation count). N~12-20.
- **(d))** Feasible if you have eye-tracking; weaker without it.
- **(e))** ISMAR/IEEE VR poster; full paper needs eye-tracking + clean effect.
- **(f))** Needs eye-tracking (see section 5).
- **(g))** Direct — tells you how to draw the highlight boxes.

Idea 3 - Phone vs head-mounted for novice hazard recognition

- **(a))** On lightweight monocular glasses (Spectacles), hands-free guidance improves novice hazard recognition where the heavy stereo HMD in Camacho-Fidalgo et al. degraded it — i.e. the failure is form-factor-specific.
- **(b))** Directly extends the N=60 paper, which the authors scoped to heavy HoloLens-class hardware.
- **(c))** Between/within subjects: phone vs Spectacles; DVs hazard recognition, accident-proxy errors, time, workload. Must be de-energized/mock equipment.
- **(d))** Moderate — Spectacles dev (Lens Studio) is real work plus a credible hazard task.
- **(e))** ISMAR/IEEE VR — their wheelhouse. Poster realistic; full paper if N and effect are strong.
- **(f))** Uses your Spectacles. No purchase strictly required.
- **(g))** Validates your “hands-free next” roadmap with data — strong for the deck.

Idea 4 - Frozen-frame vs live overlay for novices

- **(a))** Freezing the frame reduces novice cognitive load and improves step accuracy for fine inspection, at an acceptable cost to spatial grounding.
- **(b))** Nobody has cleanly isolated freeze-vs-live for novice guidance; usually live is assumed better.
- **(c))** Within-subjects, freeze vs live, N~16; DVs accuracy, time, NASA-TLX.
- **(d))** Feasible but the narrowest contribution — risk of “so what.”
- **(e))** Poster/LBW; thin for a full paper.
- **(f))** None.
- **(g))** Direct — justifies (or kills) your core UX decision.

Idea 5 - Field-photo HVAC safety-calibration benchmark

- **(a))** A real-field-photo benchmark + calibration method that fixes VLMs being “too lax on intact panels.”
- **(b))** This is an ML/dataset contribution, not XR.
- **(c))** Build dataset, measure VLM safety calibration, propose a fix.
- **(d))** Feasible and useful for the startup.
- **(e))** A CV/ML workshop, not an XR venue.
- **(f))** None.
- **(g))** Maximal product benefit; minimal XR-publishability.

4. The honest bar (per idea)

Idea	Full paper VR/ISMAR	Poster/LBW	CHI/UIST?	Not realistic?
1 Misplaced Trust	Only with strong effect + N~30-48 + clean intervention	Very realistic	CHI is the best home	-
2 Cue design	Possible with eye-tracking + clear effect	Realistic	ISMAR/VR fine	Full paper unrealistic w/o eye-tracking
3 Phone vs HMD	Stretch but plausible (extends known result)	Realistic	Stay at VR/ISMAR	-
4 Freeze vs live	Too thin	Realistic	Maybe UIST/CHI LBW	Full paper not realistic
5 Benchmark	Wrong venue	No	No (ML/CV venue)	Not an XR contribution

Reality check for a solo high-schooler: aim poster / Late-Breaking-Work first. A poster at IEEE VR or ISMAR is a real, citable, peer-reviewed XR credential. A full paper from a solo minor with no co-authors and a first study is rare. Idea 1 has genuine full-paper potential *at CHI* if the over-reliance effect is strong — treat that as the upside case, not the plan.

5. Hardware / software recommendations

Buy only what a chosen idea needs.

- **Idea 1 (recommended):** nothing required. Phone + laptop. Optional eye-tracking strengthens it.
- **Eye-tracking (Ideas 1-2):** the one meaningful purchase. Screen-based Tobii Pro Spark/Nano (~$5-7k) or refurbished Tobii; budget path = webcam gaze (GazeRecorder / WebGazer.js), free but low accuracy (poster-only); wearable Pupil Labs Neon (~$5-7k). **Honest rec:** for a poster, skip eye-tracking and use hazard-detection accuracy + NASA-TLX + forced-choice behavioral measures (objective, free).
- **Idea 3:** your Snap Spectacles suffice; Lens Studio is free. No purchase.
- **EEG (cited in Mahmood et al.):** do not buy. Out of scope; consumer EEG won't pass reviewer scrutiny.

6. Recommended path

Build Idea 1: “Misplaced Trust — when AI-driven AR safety guidance is wrong, do novices catch it?” Best balance of: genuinely novel (nobody has safety-grounded over-reliance on an imperfect AR safety AI), realistic for a solo minor (no new tracking, no required hardware), and maximally aligned with your product (the intervention you test is a feature FieldLENS needs).

Why not the head-mounted version: Idea 3 sounds more “AR,” but adds Spectacles engineering risk and an IRB problem (people moving around equipment). Idea 1 runs on a screen with static images, making the ethics and safety tractable for a minor — the single biggest constraint.

Honest constraints (don't skip)

- **You're a minor running human-subjects research.** You need an adult co-PI / supervisor (teacher, university mentor, or science-fair affiliated IRB). Top venues and ISEF require documented ethics review. Start this first — it gates everything.
- **No live electricity, ever.** Use photographs / de-energized mock panels. Better science (controlled stimuli) and removes liability.
- **Recruiting:** adult HVAC novices (or general adults as proxy novices) via a trade school, makerspace, or community college. Be honest about proxy-novice limitations.

Rough 6-month plan (now -> IEEE VR 2027 poster, Aug 31, 2026)

Month	Milestone
1	Lock an adult supervisor + IRB/ethics path. Finalize research question + hypotheses. Literature read (build on cites here).
2	Build stimulus set: ~30-40 expert-labeled HVAC photos (Safe/Caution/Do-Not-Touch). Run FieldLENS; record where the AI is wrong — that's your free, real error injection.
3	Build study app: 3 conditions — (i) verdict only, (ii) verdict + confidence, (iii) verdict + forced manual-check. Pilot with 3-5; fix protocol.
4	Run the study. Target N~24-32 within-subjects (24 = credible poster minimum; 30-48 for full-paper ambition). DVs: false-“safe” acceptance, hazard detection, trust, NASA-TLX, time.

Month	Milestone
5	Analyze (paired tests / repeated-measures ANOVA). Write. Figures. Headline hunted: the intervention cut over-reliance on wrong verdicts by X%.
6	Polish; have supervisor + an XR researcher read it. Submit to IEEE VR 2027 poster (or CHI 2027 LBW, Sep 10).

- **Baseline:** condition (i), the plain AI-verdict overlay (FieldLENS as it is today). Intervention conditions are the contribution.
- **Primary measure:** rate at which novices accept a *wrong* “Safe” verdict (the safety-critical error). Secondary: hazard-detection accuracy, trust, workload, time.
- **N:** 24 minimum for a poster; aim ~30 to give a full paper a chance.

Two caveats to be straight about

- **Novelty goes stale fast** — Guided Reality and JARVIS are 2025-26. Re-survey closest prior work right before you submit; “first to study over-reliance on imperfect AR safety AI” needs a final check.
- **The N=60 safety result is form-factor-specific** (heavy stereo HoloLens, not phones). Don't over-claim “AR is unsafe.” Frame it precisely: AI-error-induced over-reliance, which is your actual contribution.

Generated from the deep-research workflow (107 agents, 25 sources, 21 verified claims). Preprints flagged inline. Deadlines verified live June 2026 — re-confirm on venue sites before planning.